

**DONBOSCO INSTITUTE OF MANAGEMENT STUDIES AND
COMPUTER APPLICATIONS**

Kumbalagodu, Mysore Road, Bengaluru-560074

A Project Synopsis

ON

COMPARE2SAVE

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BANGALORE UNIVERSITY (JNANABHARATHI)

UNDER THE GUIDANCE OF

Ms. K Keerthi Yadav Asst. professor,

Dept. of BCA, DBIMSCA.

SUBMITTED BY

MANOJ N
(U03CQ23S0025)

ABSTRACT

The COMPARE2SAVE is a comprehensive, full-stack marketplace and local product discovery application designed to bridge the gap between neighborhood brick-and-mortar retail stores and modern e-commerce. The project is designed with a Flask-based backend in Python, using Flask-SQLAlchemy for database operations and structured SQLite/PostgreSQL databases. The frontend is built using HTML5, Vanilla CSS, and modern JavaScript (ES6) with specific integration of Three.js for an interactive 3D product visualizer and Leaflet.js for geolocation-based store mapping. The platform delivers dedicated dashboards with Role-Based Access Control (RBAC) for Customers, Shopkeepers, and Owner Admins.

Traditional shopping methods for local stores rely heavily on physical visits and manual price comparisons, which lead to high search friction, wasted travel time, and operational inefficiencies for both buyers and sellers. The proposed system replaces these outdated, manual routines with a location-aware digital catalog and direct chat platform. Customers can register, verify their mobile numbers securely using Twilio SMS OTP, search for products near their current coordinates, compare prices, and negotiate with shopkeepers via a real-time chat interface.

A core feature of the system is the integration of Three.js to provide an immersive 360-degree 3D preview of products, giving local shops an interactive edge similar to high-end e-commerce applications. Furthermore, the incorporation of an OpenAI-powered AI Shopping Assistant provides immediate recommendations and handles natural-language product discovery queries. Overall, the project aims to improve local trade, reduce consumer search overhead, and offer cross-platform installation as a Progressive Web App (PWA) and Capacitor-based mobile application.

INTRODUCTION

In recent years, the retail market has experienced a significant shift towards large e-commerce platforms. While national-scale online retail provides convenience, it lacks instant local fulfillment, and neighborhood shopkeepers face immense difficulty in making their inventory visible to nearby consumers. Conversely, local consumers who need items immediately often walk from store to store checking stock and comparing prices, which is highly inefficient.

Traditional local commerce is highly fragmented and lacks a unified platform where inventory and pricing are transparent. Small shopkeepers often do not have the technical skills or budget to maintain individual online portals, making them invisible in the digital economy.

The COMPARE2SAVE application is developed to overcome these challenges. By offering a unified marketplace, the platform enables local businesses to catalog their products easily, while giving neighborhood customers a live search engine to locate items and compare prices nearby. This system provides interactive 3D rendering, live geographical maps, role-based controls, and direct seller-to-buyer messaging, providing an immediate online-to-offline retail bridge.

OBJECTIVES OF THE PROJECT

The primary objectives of the COMPARE2SAVE system are as follows:

- To develop a complete web-based local product discovery and marketplace system using modern technologies.
- To automate inventory management for local shopkeepers and reduce search overhead for consumers.
- To implement secure, role-based user authentication using JWT and Twilio SMS-based OTP verification.
- To integrate an interactive 3D product viewer using Three.js for 360-degree item visualization.
- To incorporate Leaflet.js interactive maps to locate nearby shops and track orders geographically.
- To embed an OpenAI-powered AI Shopping Assistant for natural-language product search and recommendations.
- To establish real-time live chat communication between customers and shopkeepers.
- To provide a scalable platform that supports desktop installation (.exe), PWA capabilities, and mobile builds.

MODULES OF THE PROJECT

The COMPARE2SAVE application consists of multiple interconnected modules designed to provide a comprehensive local marketplace:

1. **User Authentication Module:** Handles secure registration, Twilio SMS OTP verification, and secure login with Role-Based Access Control for Customers, Shopkeepers, and Admins.
2. **Location & Mapping Module:** Integrates Leaflet.js to pinpoint shop locations, track current user coordinates, and compute distances between buyers and sellers dynamically.
3. **Product Discovery & Comparison Module:** A core engine that allows users to search products, sort them by price or distance, and visually inspect them. It bridges the offline-to-online gap.
4. **3D Visualization Module:** Uses Three.js to provide customers an interactive 360-degree preview of select products, enhancing the decision-making process before physical visits.
5. **AI Shopping Assistant Module:** Integrates the OpenAI GPT engine to interpret natural language queries, providing immediate product suggestions and smart recommendations.
6. **Live Communication Module:** Offers a real-time chat interface connecting buyers directly to sellers for price negotiations, stock confirmations, or general inquiries.
7. **Shopkeeper & Admin Dashboard Modules:** Provides shopkeepers tools to manage their digital catalogs seamlessly. Admins monitor transactions, user activities, and platform integrity.

EXISTING AND PROPOSED SYSTEM

EXISTING SYSTEM

Existing local commerce operates primarily through manual routines. When consumers want to purchase a product immediately, they must visit local shops physically or make phone calls to inquire about availability and pricing. Global e-commerce platforms cannot fulfill orders instantly and lack neighborhood-level store discovery. Limitations include high search costs, frequent stock unavailability issues, lack of online visibility for small sellers, and no interactive or live negotiation platform.

PROPOSED SYSTEM

The proposed COMPARE2SAVE system provides an automated, location-aware, and highly interactive marketplace solution. Built using a robust Flask backend and a modern HTML5/CSS Vanilla UI, it incorporates real-time geolocation mapping, 3D item visualization, and AI assistant capabilities. The system seamlessly handles digital product cataloging and enables location-based store discovery with direct seller-to-buyer interactions, significantly improving the local retail experience.

SOFTWARE & HARDWARE REQUIREMENTS

SOFTWARE REQUIREMENTS

- Operating System: Windows 10/11, Linux, or macOS.
- Frontend Technologies: HTML5, CSS3 (Vanilla), JavaScript (ES6), Three.js, Leaflet.js.
- Backend Technologies: Python 3.10+, Flask, Flask-SQLAlchemy, Flask-Login.
- Database: SQLite (Development) or PostgreSQL (Production).
- Additional APIs & Tools: Twilio SMS API, OpenAI API, WebSockets, PyInstaller, Capacitor.
- Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge.

HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or equivalent (or higher).
- RAM: 4GB minimum (8GB recommended for concurrent services).
- Storage: At least 5GB of free space for project dependencies and localized databases.
- Internet Connection: Stable connection required for AI API, mapping, and OTP features.
- Peripherals: Standard monitor, keyboard, mouse. Mobile devices for testing the Capacitor app.

CONCLUSION

The COMPARE2SAVE project presents a highly interactive and scalable solution that digitizes local commerce. By combining Flask backend capabilities with advanced frontend tools like Three.js and Leaflet.js, it creates a powerful platform that connects offline retail with online consumers in real-time. The implementation of Twilio OTP ensures a secure and verifiable authentication flow, while the AI assistant handles search queries intuitively. With compilation options ranging from PWAs to standalone executables and mobile packages, the platform achieves high accessibility and user convenience.

Ultimately, the application supports local economies by redirecting online consumers to nearby physical shops, laying down a stable foundation for future enhancements like augmented reality shopping experiences and automated route optimization for hyperlocal deliveries.

REFERENCES

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